UNIVERSITI TUNKU ABDUL RAHMAN
FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

UCCD3243 SERVER-SIDE WEB APPLICATIONS DEVELOPMENT

Session 202602
February 2026 Trimester

Group Assignment

(100 marks)

Deadline: 17th April 2026 (Friday), 5:00 PM

This assignment is worth 15% of the overall assessment of the course.

Assignment Question:

This **group assignment** requires developing a web application utilizing **PHP** with the integration of **relational database** where the application will facilitate the **3-tier architecture** web application. This assignment carries a total of 100 marks and contributes 15% to the overall assessment of this course.

Project Background:

You must design and implement a web application based on the following specification:

- **Student Co-curricular Management System** is a web-based application that allows students to record and manage their co-curricular activities such as event participation, club membership, merit contribution hours, and achievements in one centralised system. The application is developed using PHP and MySQL with a three-tier architecture and includes secure user registration, login, and full CRUD functions for each module. All data is linked to individual users through a shared database, enabling students to track their involvement in a structured and organised manner.
  The web application should cover the following **FOUR (4)** modules:

  1. **Event Tracker Module**
     - Focus: Formal, organised programmes.
     - This module allows users to record programmes and events they participate in, including university events, competitions, workshops, or talks. Each record represents a programme with its own event details.
     - *Basic Features: Add, view, edit, and delete event records.*

2. **Club Tracker Module**
   - Focus: Membership and roles in clubs or societies.
   - This module allows users to manage their involvement in clubs or societies by recording membership information and leadership roles. Each record represents a student's affiliation with a specific club.
   - *Basic Features: Add, view, edit, and delete club records.*

3. **Merit Tracker Module**
   - Focus: Co-curricular contribution hours.
   - This module allows users to record contribution hours completed through co-curricular involvement, such as volunteering or service activities. Each record reflects the amount of contribution within a given period.
   - *Basic Features: Add, view, edit, and delete merit records.*

4. **Achievement Tracker Module**
   - Focus: Results and recognition.
   - This module allows users to store records of achievements such as awards, certificates, and recognitions. Each record represents achievement received by the user.
   - *Basic Features: Add, view, edit, and delete achievement records.*

- ***Functional Requirements:*** Students must design and develop the system using PHP and MySQL with database connectivity and full CRUD functions for all four modules. The application must include session-based access control, cookie usage, error handling, and a relational database design that links module records to the authenticated user. A main menu or dashboard must be provided for consistent access to all modules.

  Centralized User Management:

  - The system must provide a single registration and login mechanism for all users.
  - User authentication must be handled centrally and shared across all modules.

- Optional Feature Suggestion:
  - Admin module to view all registered users and basic usage summaries, such as total events, clubs, merit records, and achievements recorded by each user.
  - Additional navigation features, such as sorting, filtering, or searching, to improve record browsing.

<u>Assignment group</u> – 4 members in a group.

There ***<u>must be 4 members</u>*** in a group. Not allowed to have 1, 2 or 3 members in a group even the member(s) is willing to work on all the required modules.

- Each member MUST choose one module to work on as individually, ensuring equal distribution of responsibilities among group members.
- Integration work between modules should be group effort and not an individual effort.
- A group leader has to be elected for each group, and the leader is expected to coordinate with other group members and to take charge at all costs in completing the assignment work. Any issues (related to management) that crop up among group members are collective responsibility of all the members to solve it amicably.

Important Notes:

1. You must hand in the soft copy of your work. You will be expected to demonstrate *(*presentation – depends on the availability of time/updating your assignment progress in the lab)* your work from the soft copy. The demonstration will be used to:
   - assess both the work and your understanding
   - ensure that the work is all your own

   Failure to be able to answer questions relating to the work will result in a loss of marks; you must attend presentation session to explain your design and implementation.

   The following software must have been used to develop the desired web application:

   1. XAMPP
   2. MySQL phpMyAdmin

   Maximum marks will only be awarded for a **well-designed, fully implemented system and well-tested.** Marks for design and documentation are included in the mark's allocation table, and they must be completed to achieve a good mark.

2. **You must include the following items in your report (follow the order given below):**

   1. A title page showing the unit code and title, your course, your name, practical group and assignment title. You also need to include the marking scheme at the front page of your report.

   2. Report length should not exceed 30 pages, excluding the cover page and references.

   3. The **content of the report of assignment** would include the following areas. *(Marks will be deducted if your report is similar with others):*
   Each section must provide details for both the overall system and individual modules where applicable.

      - *Site Hierarchy and Navigation (For Overall System)*
        - *Students are expected to prepare site map or website hierarchy diagram to show the organization structure of the website pages and illustrate the flow of navigation.*
        - *This should reflect the overall system structure, including all modules.*
        - *Include explanation describing how users move between pages/modules.*

      - *System Flowcharts (For Individual Module)*
        - *Students are expected to prepare system flowcharts for the specific modules that have been implemented in the system.*
        - *The usage of notations in the system flowchart must be accurately specified.*
        - *Include explanation of how the process works and the purpose of each step in your flowchart.*

- *Overview of Database Structure (For Overall System)*
  - *Students are expected to prepare an overview of the database structure using Database Schema Diagram or Entity Relationship Diagram to visually represent all tables, attributes, and relationships within the system.*
  - *Include explanation of how the tables are related and how user-specific data is stored and retrieved across modules.*

- *Functional Requirements (For Individual Module)*
  - *Students are expected to outline the core features and functionalities of the system.*
  - *The functionalities should align with the modules implemented.*
  - *Include explanation for each feature and how it supports the module's purpose.*

You have to submit the softcopy of your developed web application files with the report.

		Student 1	Student 2	Student 3	Student 4
STUDENT ID					
STUDENT NAME					
PROGRAMME					
MODULE					
Attribute	**Actual Marks**				
Report Section (Group)	**30**				
Site Hierarchy and Navigation	5				
System Flowcharts	10				
Overview of Database Structure	5				
Functional Requirements	10				
Application Section (Individual)	**35**				
Module Implementation	15				
Data and Logic Flow	10				
Coding Structure	5				
Error Handling and Debugging	5				
Application Section (Group)	**35**				
Professional Outlook (Front and Back End)	5				
Integration of Modules	20				
Testing and Debugging	10				
Total	**100**				

3. Format of report: IEEE Referencing System

4. Deliverables

You are required to submit the softcopy of the following items:

- Documentation of the system (**Report**).
- A softcopy of the program coded in PHP. The web application should include the following:
    - The working codes (all source files) of the entire web application should be saved in one folder.
    - Import your latest database and save the .sql file (Create a folder named **database** and place the .sql file in this folder)
    - Both the **source code folder** and the **SQL database file** should be included in a single zip file for web application deployment.

5. Submission & Plagiarism Policies

The assignment submission deadline will be on **17 April, Friday at 5pm**. If, for whatever reasons you are unable to make the date, a written explanation is required with a substantial penalty on obtained marks will be imposed. If there is no written explanation from the group leader exceeding 24 hours from the actual submission date, assignment would be rejected.
All work must be original and if, taken from any works other than yours must be properly referenced using **IEEE Referencing System**.

Submission link:
https://forms.gle/5uH45KbmCuaqtrYx5

Group leaders must submit the assignment report using the Google Form submission link and include a Google Drive link in the form that contains all related assignment materials. The Google Drive link must be publicly accessible so that lecturers can download the files.

- Only one submission per group is required. Please ensure that all files are complete and accessible before submission.